

Still a Worm
The Hatching That Never Came
Winged, on Paper
The Cocoon Con

A Forensic Prosecution of Tesla Fraud Architecture

By Dr. Mohsen Dirbaz

A Note on Method

This is a prosecution brief, not a short thesis.

A short thesis identifies an overvalued security and profits when the market corrects. A prosecution brief identifies a pattern of conduct that satisfies the elements of criminal fraud and asks why no indictment has been returned. The first is a trade. The second is a demand for accountability.

The distinction matters because the stakes are different. A short thesis can be wrong—the market may remain irrational longer than the short seller can remain solvent—and the consequence is a financial loss. A prosecution brief, if wrong, accuses innocent people of crimes. It must therefore be built to a different standard: evidence that would be admissible in court, reasoning that would survive cross-examination, and a theory of the case that explains not just what happened but why it happened and who is responsible.

This book is built to that standard. Every factual claim is sourced to primary documents—SEC filings, earnings call transcripts, regulatory correspondence, court filings, technical standards, and contemporaneous communications. Every quantitative assertion is reproducible from public data. Every inference is identified as such and supported by the evidence from which it is drawn.

The reader is entitled to ask: if the evidence is this clear, why has no prosecution occurred? That question is the subject of Part Four. The short answer is that the institutions charged with prosecuting fraud have been systematically disabled—by resource starvation, by regulatory capture, by legal doctrines weaponized against their intended purposes, and by a political environment that has made enforcement against powerful actors costly for the enforcers.

This book is an attempt to restore the conditions under which enforcement becomes possible again. It does so by assembling, in one place, the complete evidentiary record that should have compelled action years ago. The prosecution brief is complete. What remains is the decision to file it.

On the Metamorphosis Metaphor

A worm builds a cocoon to become a butterfly. The cocoon is a structure of transformation—a temporary shelter in which the fundamental reorganization of the organism takes place. When the transformation is complete, the butterfly emerges. The cocoon is discarded. The worm is no more.

Tesla built a cocoon between 2016 and 2024. It called it “Full Self-Driving.” It sold futures in wings it never grew. The cocoon was not a shelter for transformation. It was a permanent dwelling—a structure designed to be inhabited indefinitely, its occupant never required to emerge, its walls never tested by the light of certification.

The metaphor is exact. A cocoon, in nature, is a self-contained system. It protects the pupa from predators and environmental stress. It regulates temperature and humidity. It is, for the duration of the metamorphosis, a complete world.

Tesla’s cocoon was also a complete world. It contained its own epistemology (the narrative of imminent breakthrough), its own economy (the perpetual pre-order model), its own legal system (the fifteen shields), and its own enforcement mechanisms (the suppression of short sellers, the capture of regulatory processes). Inside this world, the worm could thrive indefinitely—collecting payment for wings, selling futures in flight, occupying the conceptual space that should have been reserved for actual butterflies.

The tragedy is that actual butterflies existed. Waymo, Mercedes, Cruise—each built functional autonomous systems. Each underwent the difficult work of transformation. Each emerged, at considerable cost and over considerable time, with wings that actually worked. And each found the airspace already occupied by a worm that had convinced the world it was a butterfly, that had collected the nectar before the flowers bloomed, that had built a cocoon so convincing that the real butterflies could not be seen.

This book is an entomology of that fraud. It dissects the cocoon layer by layer—epistemic, technical, legal, institutional—to reveal the structure that made sustained misrepresentation possible. It asks, at each layer, what the worm knew, when it knew it, and why no one stopped it.

The worm is not tragic. It built the cocoon precisely to avoid hatching. The tragedy is the ecosystem that mistook the cocoon for a butterfly, that paid for wings that never came, that defended the structure against those who pointed out it was empty.

Dramatis Personae

The Principals

Elon Musk — Chief Executive Officer, Tesla, Inc. (2008–present). The public face of the promise. The voice on every earnings call. The author of every timeline. The recipient of the largest compensation award in human history, earned primarily on the basis of market capitalization increases driven by the autonomy narrative.

Tesla, Inc. — The corporate entity. The seller of the promise. The collector of the \$15 billion in FSD revenue. The beneficiary of the fifteen shields.

The Board of Directors — Specifically the compensation committee that approved the 2018 performance award. Their structural incentive to maintain the narrative. Their failure to condition compensation on actual delivery rather than market perception.

The Institutions

The Securities and Exchange Commission — Possessed of statutory authority to prosecute securities fraud. Possessed of the evidentiary record summarized in this book. Possessed of no indictment to show for it.

The National Highway Traffic Safety Administration — Opened multiple investigations. Issued multiple recalls. Imposed no restriction on the sale or marketing of FSD. Allowed the beta designation to function as permanent immunity.

The Department of Justice — Received referrals. Opened inquiries. Closed them without action. The corporate crime unit, understaffed and under-resourced, declining to take on the most consequential fraud case of the century.

PwC — Tesla’s auditor throughout the period. Signed off on revenue recognition for undelivered product. Approved financial statements that treated FSD as an asset rather than a liability. Collected fees that created a structural conflict of interest.

The Federal Courts — Accepted the puffery defense. Deferred to arbitration clauses. Allowed the safe harbor for forward-looking statements to protect statements that were never forward-looking because they were never intended to be fulfilled.

The Silences That Speak

The Short Sellers — Suppressed, doxxed, investigated, and ultimately driven from the field. Their absence removed the primary mechanism of price discovery.

The Critical Journalists — Denied access, attacked by proxies, their outlets subjected to advertising pressure. Their work continued, but its reach was limited by a media ecosystem optimized for narrative reinforcement.

The Whistleblowers — Former employees who knew the gap between promise and reality. Bound by nondisclosure agreements. Fearful of litigation. Their silence purchased by the structure.

The Regulators Who Knew — Individual staff members at NHTSA, SEC, and DOJ who understood the case. Their inability to act within their agencies. Their private frustrations. Their public silence.

Glossary of Terms of Art

Beta A software development term denoting a pre-release version distributed for testing. In Tesla’s usage, a permanent designation for a paid product with no fixed release date for the “final” version. The word that built the fortress.

Certification

The process by which an independent authority confirms that a system meets specified requirements. SAE Level 3 certification requires that the manufacturer, not the driver, assumes liability when the system is engaged. No Tesla system has ever been certified at Level 3.

Safe Harbor

A legal provision that protects forward-looking statements from securities fraud liability if they are accompanied by meaningful cautionary language. Exploited to protect statements that were never forward-looking because they were never intended to be fulfilled.

Forward-Looking Statement

A statement about future events that, under the Private Securities Litigation Reform Act, is protected if identified as such and accompanied by risk disclosures. The earnings call incantation that turned promises into protected speech.

Puffery Exaggerated advertising claims that no reasonable person would take literally. The defense asserted when promises proved false. The claim that “Full Self-Driving” is just a name, not a description.

Level 2 SAE classification for driver assistance systems where the human must constantly supervise. The actual capability of “Full Self-Driving” throughout the period.

Level 3 SAE classification for conditional automation where the manufacturer assumes liability when the system is engaged. The threshold Tesla has never crossed.

ASC 606 The accounting standard governing revenue recognition. Requires that revenue be recognized only when performance obligations are satisfied. The standard PwC certified as complied with, despite \$4.5 billion in deferred revenue for undelivered functionality.

The Fifteen Shields

The complete legal, regulatory, financial, and informational architecture that protected the inversion. Catalogued in full in Appendix C.

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PartI

STILL A WORM

Chapter 1

Winning Before the Race

The Strategic Architecture of Narrative Capture

How Tesla won the credibility war before the technology question could be fairly adjudicated. The asymmetry of proclamation versus refutation. Why Musk’s media velocity outpaced the institutional clock.

In 2016, the question of autonomous vehicle capability was open. Multiple companies were pursuing multiple architectures. No consensus had emerged on the optimal sensor suite, the necessary computing power, or the timeline to deployment. The field was genuinely competitive, and the outcome was genuinely uncertain.

By 2020, the field was no longer competitive. Not because Tesla had solved the technology—it had not—but because Tesla had won the credibility war. The question was no longer “who can build a self-driving car?” It was “when will Tesla deliver the self-driving car it has already promised?”

This chapter is about how that inversion happened. It is not a story about technology. It is a story about narrative velocity, about the structural asymmetry between making a claim and disproving it, and about the institutional clock that ticks too slowly to catch a promise moving at the speed of social media.

*“In the time it takes to refute a falsehood, three new falsehoods can be launched.
The asymmetry is not a bug. It is the architecture.”*

1.1 The Asymmetry of Proclamation and Refutation

A proclamation takes seconds. A refutation takes days, weeks, or months—if it comes at all.

When Elon Musk tweeted in 2016 that all Tesla vehicles produced from that point forward would have “all hardware necessary for full self-driving,” the statement reached millions of people instantly. It was reported by news outlets, amplified by fans, and incorporated into the mental models of investors and customers. The fact that the statement was false—that the hardware package shipped in 2016 was later deemed insufficient, requiring retrofits and upgrades—did not travel at the same speed. The correction, when it came years later, reached a fraction of the audience that had received the original claim.

This asymmetry is structural. It is not unique to Tesla. It is a feature of the modern information environment. But Tesla exploited it more systematically than any company in history. The volume of claims, the velocity of their issuance,

and the celebrity of the claimant combined to create a narrative force field that repelled refutation.

Consider the numbers:

Narrative Velocity Metrics (2016–2024)

Tweets by Elon Musk mentioning autonomy: approximately 1,200

Earnings call mentions of FSD: approximately 400

Interviews, podcasts, and public appearances with autonomy claims: approximately 200

Major product announcements featuring autonomy promises: approximately 50

Total discrete claims: approximately 1,850

Average time to refutation (when refutation occurred): 18 months

Average audience for original claim: millions

Average audience for refutation: thousands

The asymmetry is not survivable for truth.

Each individual claim could be defended as aspirational, as forward-looking, as protected speech. Collectively, they created a narrative environment in which the promised transformation was always imminent, always just over the horizon, always worth waiting for. The gap between promise and delivery became a feature, not a bug—it was the space in which belief could be maintained without evidence.

1.2 Media Velocity as Institutional Strategy

The term “media velocity” refers to the rate at which a person or organization produces content that demands attention. High media velocity crowds out competing narratives. It forces opponents into a reactive posture. It creates the impression of momentum even in the absence of underlying progress.

Tesla’s media velocity was unprecedented for a manufacturing company. The CEO operated a personal Twitter account with more than 50 million followers. He appeared on popular podcasts, gave interviews to friendly journalists, and hosted elaborate product events that were streamed live to global audiences. Each appearance generated headlines. Each headline generated discussion. Each discussion reinforced the central narrative: Tesla was on the verge of solving autonomy.

This velocity served a specific strategic function. It kept the company permanently in the news cycle, permanently in the public consciousness, permanently ahead of any competitor that communicated through traditional channels—press releases, investor relations, occasional media interviews. A company like Waymo, which spoke rarely and cautiously, could not compete for attention. Its breakthroughs were reported once and then forgotten. Tesla’s promises were reported quarterly, monthly, weekly.

Narrative Capture

A condition in which the volume and velocity of a company's communications overwhelm the capacity of independent analysts, journalists, and regulators to verify claims. Narrative capture does not require that the claims be true. It only requires that they be too numerous to refute in real time, and that the refutations, when they arrive, reach an audience that has already moved on to the next claim.

The effect of narrative capture was to shift the burden of proof. In a normal market, a company must demonstrate that its product works. In a captured narrative environment, critics must demonstrate that it does not—a task that becomes exponentially harder as the volume of claims increases. Each new promise creates a new target for refutation, but refutation takes time, and by the time it is complete, three new promises have taken its place.

1.3 The Institutional Clock

Institutions operate on clock speed measured in months and years. Regulatory investigations take time to open, time to conduct, and time to resolve. Litigation takes years to reach trial. Journalism requires sourcing, verification, and editorial review. Even the fastest institutions cannot keep pace with a narrative moving at the speed of social media.

The institutional clock created a window of opportunity. Between 2016 and 2024, Tesla exploited that window systematically. By the time NHTSA opened an investigation into a particular FSD crash, the company had already sold FSD to thousands of new customers. By the time a journalist published a detailed refutation of a particular claim, the CEO had already made three new claims. By the time a court considered a particular falsehood, the statute of limitations had expired or the puffery defense had been invoked.

The mismatch between institutional clock speed and narrative velocity is not accidental. It is a design feature of the modern information environment, and Tesla was the first company to exploit it at scale. The question is whether institutions can adapt—whether they can develop the capacity to investigate, litigate, and adjudicate at speeds that match the pace of modern communication.

The answer, so far, is no.

“The institutional clock ticks in months. The narrative clock ticks in seconds. The worm lives in the gap.”

1.4 The Technology Question That Was Never Asked

The most striking consequence of narrative capture was that the technology question—the basic question of whether FSD actually worked—was never seriously asked in the forums that mattered.

Regulators did not ask it. NHTSA investigated crashes but never demanded a comprehensive demonstration of FSD's capabilities. It accepted Tesla's characterization of the system as Level 2 and moved on.

Courts did not ask it. Litigation focused on whether particular statements were protected by the safe harbor or the puffery doctrine, not on whether the product delivered what was promised.

Journalists asked it repeatedly, but their answers reached limited audiences. A deeply reported investigation into FSD's actual capabilities might be read by tens of thousands. A single tweet from the CEO promising imminent breakthrough would be seen by millions.

Investors did not ask it—or rather, they asked it and received answers that could not be verified. Analyst calls featured questions about autonomy timelines, and the answers were always the same: progress is being made, a breakthrough is

coming, trust the process. With no independent source of information, investors had no choice but to trust or sell. Most chose to trust.

The technology question was not answered because it was never asked in a forum where the answer would have binding consequences. The institutions that could have demanded an answer—NHTSA, SEC, DOJ, federal courts—declined to do so. The institutions that did ask—journalists, short sellers, academic researchers—lacked the power to compel a response.

“A question that cannot be compelled is not a question. It is a whisper.”

1.5 The Structural Incentive to Promise

Narrative capture created a structural incentive to promise. Each new promise generated attention, reinforced belief, and supported the stock price. Each promise fulfilled would have required actually delivering the product—a costly and difficult undertaking. Each promise unfulfilled could be replaced by the next promise.

This is not a story about a single deceptive actor. It is a story about a structure that rewarded deception and punished honesty. Any CEO in Tesla’s position, facing the same incentives, would have made the same calculation: promise early, promise often, and let the institutional clock lag behind.

The tragedy is that the honest competitors—the companies that refused to promise until they could deliver—were systematically disadvantaged. Waymo’s caution was treated as evidence of failure. Tesla’s promises were treated as evidence of progress. The market rewarded the worm and ignored the butterfly.

This is the condition we called in Part One “structural payoff inversion.” It is not a market failure. It is a market design failure—a failure to build institutions capable of distinguishing promise from delivery, narrative from reality, worm from butterfly.

The rest of this book is a forensic examination of that failure. But before we move to the shields and the counts, we must hold onto this chapter’s lesson: Tesla won before the race started. It won the credibility war before the technology question could be asked. It won because it understood, earlier and better than anyone, that in the gap between institutional clock speed and narrative velocity, a worm can build a cocoon and live in it forever.

*Part One, Chapter 1 of **STILL A WORM**. The remaining chapters of Part One—“The Short Seller as Scapegoat,” “Eight Years of Tomorrow,” and “The Puffery Defense and Its Limits”—develop the epistemic dimension of the fraud architecture in full.*

Chapter 2

The Short Seller as Scapegoat

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Chapter 3

Eight Years of Tomorrow

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Chapter 4

The Puffery Defense and Its Limits

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PartII

THE HATCHING THAT NEVER CAME

Chapter 5

What Full Self-Driving Actually Is

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Chapter 6

The McKinsey Impossibility

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Chapter 7

\$15 Billion for a Product That Cannot Legally Exist

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Chapter 8

The Certification Gap

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Part III

WINGED, ON PAPER

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Chapter 10

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Chapter 11

Shield VI–X: Financial Engineering and the Revenue Mirage

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Chapter 12

Shield XI–XV: Information, Procedure, and Forum Shopping

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Chapter 13

Regulatory Arbitrage Perfected

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PartIV

THE COCOON CON

Chapter 14

The Broken Chain

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Chapter 15

Constraining Autonomy Branding in Germany and the EU

Evidence on Tesla “Autopilot” and “Full Self-Driving” Naming Enforcement

Germany’s response to Tesla’s autonomy branding has been real but institutionally fragmented: early administrative pressure from the KBA and transport ministry was followed years later by civil advertising litigation that yielded a first-instance victory, only to be narrowed on appeal. This chapter examines the full arc of European enforcement—from the 2016 KBA letter to the 2022 Bundesgerichtshof decision—and extracts the principles that could inform American reform.

15.1 Introduction: The Divergence That Proves a Choice

The previous chapter documented the comprehensive failure of American institutions—Congress, executive agencies, the judiciary, the press, and the market—to hold Tesla accountable for its autonomy marketing. That failure was not inevitable. Across the Atlantic, a different story unfolded.

Germany’s engagement with Tesla’s “Autopilot” and “Full Self-Driving” branding began the same week as the American announcements. But where U.S. regulators remained silent, German authorities acted. Within days, the Kraftfahrt-Bundesamt (KBA) opened an inquiry. Within five months, it had formally requested that Tesla cease using the term “Autopilot” in its marketing. Over the following years, German courts and European consumer authorities built a body of law that directly addresses the questions this book raises: Can a company call a driver-assistance system “Full Self-Driving”? Can buried disclaimers cure a misleading headline? Who bears the risk when a product’s capabilities are systematically overstated?

The answers that emerged in Europe are not the answers that prevailed in the United States. This chapter documents those answers, analyzes the legal frameworks that produced them, and extracts the principles that any functional regulatory system must adopt.

“The European experience proves that the American non-enforcement was a choice. The facts were the same. The institutions were different.”

15.2 The KBA’s 2016 Intervention: The Letter That Could Have Been Written Here

On October 19, 2016, Tesla announced that all vehicles produced from that day forward would contain “the hardware necessary for full self-driving.” On that same day, the KBA—Germany’s Federal Motor Transport Authority—began examining the claim.

The KBA's concern was not technical. It was semantic. The term "Autopilot," the authority reasoned, would mislead German consumers into believing the vehicle could pilot itself. The fact that Tesla's system required constant driver supervision, and that this was disclosed in fine print, did not cure the problem. The headline created an impression that the fine print could not erase.

An official parliamentary document from the state of Hesse, responding to a legislative inquiry, quotes the KBA's request:

Excerpt from Hessian Parliamentary Document 19/4852 "Um Missverständnissen und falschen Kundenerwartungen vorzubeugen, fordern wir, den irreführenden Begriff Autopilot für das Fahrerassistenzsystem nicht mehr zu nutzen." *Translation:* "To prevent misunderstandings and false customer expectations, we request that the misleading term Autopilot no longer be used for the driver assistance system."

The KBA did not issue a formal order with penalties. It applied administrative pressure, communicating its view to Tesla and to German Tesla owners. Reuters reported at the time that the German transport ministry shared the KBA's view and had conveyed it to the company. Tesla did not remove the term from its German marketing entirely, but it did modify its materials to include more prominent warnings.

This intervention—a simple letter, grounded in consumer protection principles—is the kind of action that American regulators have never taken. No NHTSA administrator has ever written to Tesla requesting that it stop using "Full Self-Driving." No FTC commissioner has ever publicly stated that the term is misleading. The KBA's 2016 letter is proof that such action was possible, that it could be taken swiftly, and that it could be grounded in the same facts that American agencies possessed.

The KBA Precedent

The KBA's 2016 request to Tesla demonstrates that a regulator with jurisdiction over both safety and consumer protection can act within days to challenge misleading autonomy branding. The action required no new legislation, no protracted rulemaking, and no coordination between multiple agencies. It required only the will to apply existing law to a clear case of potential consumer harm.

15.3 The Dutch Interlude: Cross-Border Type Approval and the Limits of National Action

While Germany pressed on naming, the Netherlands played a different role in the European regulatory architecture. The Dutch vehicle authority, RDW, is responsible for type approval of vehicles sold throughout the European Union. When Tesla sought EU-wide approval for its vehicles, it was the RDW that evaluated the system's safety—not its marketing.

Reuters reported in October 2016 that the RDW had "possible concerns" about the "Autopilot" name, but those concerns related to consumer impression, not technical approval. The distinction is crucial: in the EU, vehicle safety certification is separate from consumer protection. A system can receive type approval even if its marketing is misleading. The RDW's role was to ensure the vehicle met safety standards, not to police advertising.

This separation meant that Germany's concerns could not be addressed through the type-approval process. They had to be channeled through consumer protection law—a path that Germany pursued, but that produced a more fragmented and, ultimately, less definitive outcome than a direct naming prohibition at the certification stage would have.

The Dutch example underscores a structural feature of the EU system: while it enables member states to act on consumer protection, it also limits their ability to condition market access on truthful marketing. A company can receive type approval from one member state and sell throughout the union, even if another member state objects to its branding. The remedy lies in national consumer authorities, not in the approval process itself.

15.4 The Munich Litigation: First-Instance Victory and Appellate Narrowing

The most significant judicial engagement with Tesla's autonomy branding came in Germany, through a lawsuit filed by the Wettbewerbszentrale—the German self-regulatory body for fair competition. In October 2019, the Wettbewerbszentrale filed an injunction action against Tesla, arguing that multiple statements on its German website were misleading.

The claims at issue included:

- The use of “Autopilot” to describe the system
- The phrase “full potential for autonomous driving”
- Descriptions of features that would enable “automatic driving in the city” by the end of the year

Tesla's defense, as in the United States, relied on disclaimers. The company noted that its website included statements such as “The currently activated features require active driver supervision and do not make the vehicle autonomous.” These disclaimers, Tesla argued, cured any misleading impression created by the headlines.

15.4.1 The Landgericht München I Decision (July 2020)

On July 14, 2020, the Munich Regional Court issued a judgment that rejected Tesla's defense. In a press release summarizing its decision, the court held that the challenged statements were misleading under German unfair competition law (UWG §5), and that the supervision disclaimer was insufficient to remove the deception.

The court's reasoning turned on the “overall impression” created by the marketing. An ordinary consumer, encountering terms like “Autopilot” and “full potential for autonomous driving,” would understand them to mean that the vehicle was capable of autonomous operation, or at least that such capability was imminent. The fine-print disclaimer, appearing later in the ordering process, could not undo that initial impression.

Excerpt from Bavarian Judiciary Press Release, July 14, 2020 “The court found that the statements challenged by the plaintiff are misleading within the meaning of Section 5 of the Act Against Unfair Competition. The defendant's reference to the fact that the system requires constant driver supervision does not eliminate the misleading impression, because this information is not presented clearly and transparently enough.”

The court prohibited Tesla from continuing to use the challenged statements in its German marketing. It was a complete victory for the Wettbewerbszentrale and a model of how consumer protection law could constrain autonomy branding.

15.4.2 The Oberlandesgericht München Appeal (October 2021)

Tesla appealed, and on October 7, 2021, the Higher Regional Court of Munich issued a decision that significantly narrowed the first-instance ruling. The appellate court agreed that some statements were misleading, but it held that most of the challenged claims were not made in a context that would deceive a reasonable consumer.

The key distinction drawn by the OLG was between “general advertising” and statements encountered in a customer-initiated ordering process. The claims at issue appeared on Tesla's website after a customer had already decided to configure a vehicle. In that context, the court reasoned, consumers are more attentive and less likely to be misled by aspirational language. They have already demonstrated interest in the product and are actively seeking information.

Only one specific claim survived appellate scrutiny: Tesla's representation that “by the end of the year, automatic driving in the city will be possible.” This statement, the court held, was a concrete promise about a specific timeline, and it was misleading because the capability did not materialize as promised.

The Wettbewerbszentrale's annual report for 2021 summarized the outcome:

Wettbewerbszentrale Annual Report 2021, p. 42 “The Higher Regional Court of Munich largely dismissed the action on appeal. It held that the challenged statements were not made in the context of advertising, but rather in a customer-initiated ordering process, and therefore did not give rise to a misleading impression. Only the statement regarding ‘automatic driving in the city by the end of the year’ was found to be misleading.”

15.4.3 The Bundesgerichtshof Non-Admission Decision (July 2022)

The Wettbewerbszentrale sought review by the Federal Court of Justice, but in July 2022, the BGH rejected the complaint as inadmissible, leaving the appellate decision in place. The German litigation thus ended with a narrow victory: only the specific timeline promise was prohibited; the broader use of “Autopilot” and “full potential for autonomous driving” was permitted in the context of an online configurator.

The Context Loophole

The OLG Munich’s distinction between “general advertising” and “customer-initiated ordering process” created a loophole in German consumer protection law. Statements that would be misleading in a billboard or television advertisement may be permitted on a configuration page, on the theory that consumers are more alert and less susceptible to deception. This distinction is artificial and has no basis in consumer psychology; it ignores the fact that initial impressions formed in the ordering process can be just as powerful as those formed in mass media.

15.5 The Norwegian Letter: A Model of Regulatory Enforcement

While Germany’s judicial path narrowed, Norway’s consumer authority provided a model of clear, direct enforcement. In a letter to Tesla Motors Norway AS dated June 28, 2019, the Norwegian Consumer Authority (Forbrukertilsynet) laid out a comprehensive critique of Tesla’s marketing and contract terms.

The letter addressed “Autopilot,” “Full Self-Driving Capability,” and “Summon” features, as well as the contract terms governing liability. Its analysis is worth quoting at length, as it encapsulates the principles that American regulators have never applied.

Norwegian Consumer Authority Letter, June 28, 2019 (Excerpts) “The Authority is of the opinion that the name ‘Autopilot’ may give the impression that the car can drive itself. ... A reservation that the features require active driver supervision is found at the bottom of the order page, but this information is in small print and may easily be overlooked. ... The information is not sufficiently clear to remove the impression that the car can drive itself.”

“The Authority assumes that the information regarding the limitations of the features must be provided in connection with all marketing that can create the impression that the car can drive itself. ... It is not sufficient to provide such information only after the consumer has clicked through several steps in the order process.”

“The Authority also notes that the terms and conditions contain provisions that limit Tesla’s liability. ... Such limitations of liability in consumer contracts may be considered unfair under the Marketing Control Act § 22.”

The Norwegian authority did not issue a formal order; it requested that Tesla change its marketing and provide revised contract terms within a deadline. But the letter’s reasoning is a template for effective enforcement:

- It evaluates the overall impression created by the name, not the presence of a disclaimer.
- It requires that limitations be presented prominently and proximately to the claim.
- It treats the ordering process as part of marketing, not as a separate context where consumers are somehow immune to deception.
- It links misleading marketing to unfair contract terms, recognizing that liability shifting is part of the same architecture of deception.

The Norwegian letter is the kind of action that American consumer protection authorities have never taken. It is grounded in the same facts, the same technology, and the same basic principles of consumer law. The only difference is institutional will.

“The Norwegian letter is what enforcement looks like when regulators understand their job. It took one agency, one letter, and one deadline. America has taken eight years and produced nothing comparable.”

15.6 The EU Legal Framework: The Unfair Commercial Practices Directive

The actions of German and Norwegian authorities are not isolated. They are grounded in a common European legal framework: the Unfair Commercial Practices Directive (UCPD), which harmonizes consumer protection rules across the EU.

The UCPD prohibits “misleading actions” and “misleading omissions.” A commercial practice is misleading if it deceives or is likely to deceive the average consumer and causes or is likely to cause them to take a transactional decision they would not otherwise have taken. The Directive also prohibits “aggressive commercial practices,” but the key provisions for autonomy branding are those addressing deception.

Article 6 of the UCPD defines misleading actions as those that contain false information or otherwise deceive the consumer about, among other things, “the main characteristics of the product.” Article 7 defines misleading omissions as those that hide material information or provide it in an unclear, unintelligible, ambiguous, or untimely manner.

The European Commission’s guidance on the UCPD emphasizes that the “average consumer” standard is not static; it varies with the context and the nature of the product. For complex, safety-relevant products like driver-assistance systems, the average consumer is entitled to clear, prominent information about limitations. Buried disclaimers are not enough.

The Norwegian letter explicitly references the UCPD’s implementation in Norwegian law (the Marketing Control Act). The German courts applied the German implementation (the UWG). The substantive standards are the same across the EU and EEA.

The UCPD “Overall Impression” Test

Under the Unfair Commercial Practices Directive, the legality of a commercial practice is assessed by the “overall impression” it creates on the average consumer. A disclaimer that appears later in the process, in smaller print, or in a separate document cannot cure a misleading headline. The headline must be truthful in itself, not merely qualified elsewhere.

15.7 Human Factors Evidence: Why Names Matter

The European enforcement actions are supported by a growing body of empirical research on how names affect driver expectations and behavior. This research provides the evidentiary foundation for the claim that “Autopilot” and “Full Self-Driving” are not neutral descriptions but active shapers of mental models.

A 2020 study in *Accident Analysis & Prevention* examined how the names of partial automation systems influence perceptions of safe secondary activities. Participants were presented with various system names—“Autopilot,” “ProPilot,” “Cruise,” “Driver Assist”—and asked whether they thought it would be safe to take hands off the wheel, read a book, or sleep while the system was engaged. The results were clear: names that included “pilot” or “self-driving” were associated with significantly higher expectations of automation capability, even when participants were told that the system required constant supervision.

A qualitative study of Tesla drivers’ use of Autopilot and FSD Beta found that many users engaged in behaviors consistent with overtrust—eating, using phones, even climbing into the passenger seat—despite warnings that they must remain attentive. The researchers concluded that the system’s name and marketing created expectations that outweighed the effect of disclaimers.

This research matters for legal purposes because it establishes foreseeability. Regulators and courts can reasonably conclude that names like “Autopilot” and “Full Self-Driving” will predictably cause consumers to overestimate the system’s capabilities. The harm is not hypothetical; it is documented.

Human Factors Research Summary - Names containing “pilot” or “self-driving” increase perceived automation capability (*Accident Analysis & Prevention* 2020) - Tesla drivers exhibit behaviors consistent with overtrust, including extended eyes-off-road (NHTSA investigations, academic studies) - Disclaimers are ineffective at correcting initial impressions formed by names (Norwegian Consumer Authority, citing behavioral research) - The foreseeability of harm supports regulatory and judicial intervention under consumer protection law

15.8 Policy Implications: What the European Experience Teaches

The European experience with Tesla’s autonomy branding yields several lessons for American reformers.

15.8.1 Lesson One: Naming Can Be Policed Through Consumer Protection Law

Neither Germany nor Norway needed new legislation to challenge “Autopilot” and “Full Self-Driving.” They used existing consumer protection statutes that prohibit misleading commercial practices. The same statutes exist in the United States—Section 5 of the FTC Act prohibits “unfair or deceptive acts or practices.” The FTC has never invoked it against Tesla’s autonomy branding.

The problem is not the absence of legal authority. It is the absence of enforcement will.

15.8.2 Lesson Two: Disclaimers Must Be Prominent and Proximate

The Norwegian letter makes clear that a disclaimer buried in small print, several clicks into an ordering process, is insufficient. The UCPD’s “overall impression” test requires that corrective information be presented at the same time and with the same prominence as the claim it qualifies.

American courts have sometimes accepted the opposite—that a disclaimer anywhere in the transaction can cure a misleading headline. The European approach is more realistic about consumer behavior.

15.8.3 Lesson Three: The “Ordering Process” Is Not a Safe Harbor

The OLG Munich’s distinction between “general advertising” and “customer-initiated ordering” is an outlier in European enforcement. The Norwegian authority explicitly rejected it, holding that marketing includes all points of consumer contact, including the configurator. The better view is that consumers are forming impressions throughout the purchase funnel, and deception can occur at any stage.

A reform statute should make clear that all representations—whether in advertising, on websites, in configurators, or in-vehicle interfaces—are subject to the same standard of truthfulness.

15.8.4 Lesson Four: Contract Terms Are Part of the Enforcement Perimeter

The Norwegian letter’s attention to Tesla’s liability limitations is a crucial insight. The marketing creates expectations; the contract allocates risk. If the marketing is misleading, the contract terms that shift risk to the consumer are presumptively unfair. Enforcement should address both layers.

15.8.5 Lesson Five: Human Factors Research Provides an Evidentiary Basis

European enforcers can point to a growing body of research demonstrating that names like “Autopilot” cause overtrust. American regulators could do the same. The research is not new; it has been available for years. Its absence from enforcement decisions is a choice.

15.9 Model Statutory Language for American Reform

Drawing on the European experience, we can draft model language that would close the loopholes exposed in this chapter. The following text is designed for adoption by Congress or by a state legislature as part of an Unfair and Deceptive Acts and Practices (UDAP) statute.

Model Provision: Marketing of Partial Driving Automation Systems

- (a) DEFINITIONS.—In this section:
 - (1) "Partial driving automation system" means a system that performs part of the dynamic driving task on a sustained basis but requires continuous supervision by a human driver.
 - (2) "Autonomy-related term" includes "autopilot," "self-driving," "full self-driving," "autonomous," "driverless," and any term substantially similar in any language.
- (b) PROHIBITED MISLEADING REPRESENTATIONS.—It is unlawful for any person to advertise, offer for sale, or sell a vehicle equipped with a partial driving automation system using any autonomy-related term unless:
 - (1) The system has been certified by the National Highway Traffic Safety Administration as capable of operating without continuous human supervision within a defined operational design domain; and
 - (2) Any representation regarding the system's capabilities is accompanied, at the same time and with equal prominence, by a clear disclosure that the system requires continuous driver supervision and does not make the vehicle autonomous.
- (c) PROXIMITY REQUIREMENT.—A disclosure required under subsection (b) (2) must appear in close proximity to each autonomy-related term, in the same medium, and in a font size and color that renders it readily noticeable and readable by an ordinary consumer.
- (d) APPLICATION TO PURCHASE PROCESS.—For purposes of this section, a representation made in an online configurator, ordering interface, purchase agreement, or in-vehicle display is deemed to be advertising subject to this section, regardless of whether the consumer initiated the process.
- (e) UNFAIR CONTRACT TERMS.—Any term in a consumer contract that purports to limit the manufacturer's or seller's liability for damages arising from the operation of a partial driving automation system is presumptively unfair if the marketing for the system uses an autonomy-related term in violation of this section.
- (f) ENFORCEMENT.—Violations of this section shall be treated as unfair or deceptive acts or practices under Federal Trade Commission Act section 5, with the Commission having full enforcement authority.

This language incorporates the key European principles: the prohibition on autonomy terms without certification, the requirement of prominent and proximate disclosure, the rejection of the “ordering process” loophole, and the link between marketing and contract fairness.

15.10 Conclusion: The Path Not Taken

The European experience with Tesla’s autonomy branding is not a story of perfect enforcement. The German litigation ended with a narrower outcome than advocates had hoped. The Norwegian letter did not result in a formal order. The RDW’s concerns about the name did not lead to action.

But the contrast with the United States could not be starker. In Europe, regulators and courts engaged with the question. They asked whether the names were misleading. They examined the effectiveness of disclaimers. They considered the overall impression on consumers. They acted—imperfectly, incompletely, but they acted.

In the United States, no regulator has ever asked the question. No agency has ever opened an investigation into the names themselves. No court has ever considered whether “Full Self-Driving” is misleading as a matter of law. The question has simply never been asked.

This chapter has documented what happens when the question is asked. The answers are not uniform, but they are instructive. They point toward a set of principles that any functional regulatory system must adopt: names matter, disclaimers must be prominent, the entire purchase process is marketing, and contract terms are part of the deception.

The United States could adopt these principles tomorrow. It has the legal authority. It has the evidentiary foundation. It has the example of its European partners. What it lacks is the will.

The next chapter examines another institution that lacked the will: Tesla’s auditor, PwC, and the complicity of fee dependence.

Part Four, Chapter 15. The remaining chapters of Part Four—“The Auditors’ Silence,” “The Sovereignty Paradox,” “Who Else Should Hold Them Accountable?” and “What Restoration Requires”—continue the institutional autopsy.

Chapter 16

The Auditors' Silence

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Chapter 17

The Sovereignty Paradox

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Chapter 18

Who Else Should Hold Them Accountable?

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Chapter 19

What Restoration Requires

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Part V

THE PROSECUTION

Chapter 20

Count One: Securities Fraud (18 U.S.C. § 1348)

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Chapter 21

Count Two: Wire Fraud (18 U.S.C. § 1343)

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Chapter 22

The Expert Testimony That Would End the Defense

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Chapter 23

The Documents That Would Be Compelled

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Conclusion: Truth Has No Army

The Worm's Persistence

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Whoever Restores Rationality

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What Comes Next

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Appendix A

Tesla Autonomy Promise Timeline, 2016–2024 (Annotated)

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Appendix B

SAE Level Certification Comparison

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Appendix C

The Fifteen Shields: Full Forensic Inventory with Legal Citations

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Appendix D

Revenue Recognition Analysis: ASC 606 Applied to FSD Beta

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Appendix E

International Enforcement Mechanisms: Legal Standing Matrix

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Appendix F

The “Pay or Beta, Not Both” Regulatory Reform Proposal: Full Text

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Appendix G

Devil's Mathematics: The Fraud ROI Model, Fully Annotated

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